

Kingston Ghost Tree

Kingston's first PCIe Gen 4.0 SSD is on the horizon

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Kingston had some interesting launches around CES 2021 of which two were particularly interesting to us. We spoke with Mark Noland, SSD Technology Manager and Shealyn Johnson, Communications Specialist, Kingston to know more about the new Workflow Station dock and Ghost Tree SSD.

Q What was the idea behind the Kingston Workflow Station?

Shealyn Johnson: Everyone at this point is a content creator. No matter whether you're an influencer or perhaps you have a photography hobby or you like to share an unboxing with your family for whatever reason. Content is king right now and everyone has one or more SD Cards, USB Flash drives and other devices lying around and it's all about being able to access that content. And you want to be able to read and write to these devices at a high speed or at their advertised speeds. We don't want bottlenecks. With the Workflow Station, you're getting to your content faster and thus, to your work faster. Most people have multiple storage media and you end up carrying multiple cables and readers so we're looking at simplicity.

Q Most docks like the Kingston Workflow Station tend to come with a quick back up button that copies everything on your SD card or flash drive into predefined folders. Do you have plans to incorporate similar features into the Workflow Station?

Shealyn Johnson: There's nothing like an automatic backup or a quick backup feature on the Workflow station yet. Overall for Kingston, we're not usually the first to market

with something. We'll let the other guys put products out on the market and we'll see how our customers are using it, and we'll fine tune it and launch a product. As for back up, you can use all the ports on the Workflow Station simultaneously. So if you were out on a shoot and you were using multiple drives, then you can move data from one to another using the Workflow Station.

Q Semi-professional and professional cameras tend to use proprietary form factors such as CF (CompactFlash) and CFexpress. Do you have any plans to launch Workflow Station modules to support those interfaces?

Shealyn Johnson: CFexpress is a little newer to the market and there aren't many devices using it and it's mostly devices on the higher-end that are using it. As that becomes more mass adopted and as we find the need for it, that's when Kingston will start exploring it. For now, it's just going to be the standard SD and MicroSD modules.



Mark Noland: CF is a patented interface so anyone who has to make devices supporting it has to pay WD a licensing fee. Kingston, typically, doesn't look towards such arrangements.

Q With regards to "Ghost Tree" which is the codename for your PCIe Gen 4.0 NVMe SSD, the other manufacturers got to market a year and a half ago so was there any particular reason for Kingston to wait it out and not launch a PCIe Gen 4.0 device for so long?

Mark Noland: When we were looking to design a PCIe Gen 4.0 device, the other manufacturers had units but

if you look at the speeds that they've reached then you can see that the NAND they've used isn't capable of taking advantage of the full bandwidth of the bus. Kingston focuses on manufacturing things that we can sell a lot of. Motherboards with PCIe Gen 4.0 sockets were very limited and the technology wasn't quite there yet so there wasn't a huge demand for it. Also, if



you look at the bandwidth of the existing Gen 4.0 devices, none of them are quite up to the 7 GBps bandwidth and that's what we're aiming for. It's a really robust, high-performance gaming device aimed at prosumers. Some of the existing PCIe Gen 4.0 devices are only capable of 5.3 GBps peak performance. We're focusing on TLC NAND rather than QLC in our devices for endurance and also we're waiting for the NAND and controller technology to catch up for Ghost Tree.

Q Have you experienced any throttling during sustained reads and writes?

Mark Noland: Not really. If you have the SSD mounted with the thermal tape that's needed with the motherboard and if you have a minimal amount of air going over it then it will stay cool. If you're in a system that's locked up and there's a lot of hot air coming from the GPU, then eventually you might see throttling. Reading doesn't really heat them up but writing does. So if you were running a SQL database then that's a use case where you'd want not necessarily a heat sink but about 300 Linear Feet of air per Minute going over it. That should be enough. **Q**